

Aligning Metabolomic Networks via Graph Learning for Common Subnetwork Discovery

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June, 2026

Problem formulation

The mass spectrometry analysis includes two approach:

- **Targeted approach:** the molecule of interest or m/z values (for MS and MS/MS) are predefined for quantitation of analytes present in a complex samples matrix.
- **Untargeted approach:** is the analysis to identify the unknowns in a sample, using parent ion and product or fragment ion analysis (MS/MS).

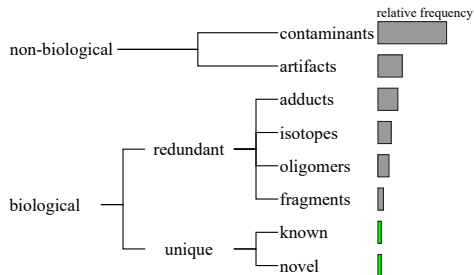


Figure 1: Composition of an untargeted metabolomic data.

Source: Adapted from [?]

Untargeted mass spectrometry workflow

A data matrix (sample vs. metabolites) acquired from a mass spectrometry-based metabolomics study includes not only biological information but also technical noise.

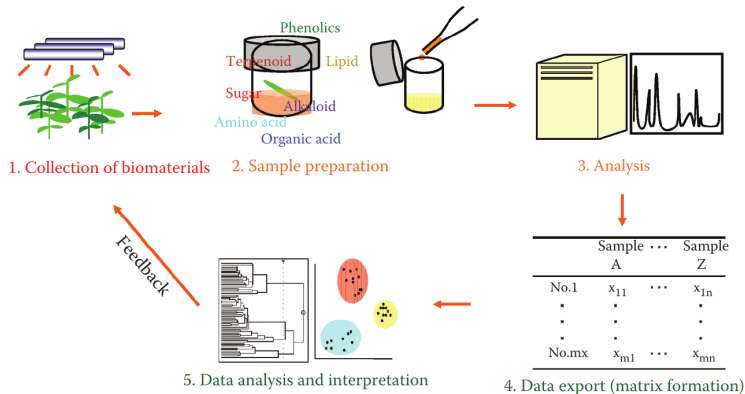


Figure 2: Metabolomics workflow.

Source: [?]

Our objectives

Aligning metabolomic networks for filter MS data.

Graph Neural Network (GNN)

It is a function $f: (A, X) \rightarrow Z$ that maps each node $v \in V$ of a graph $G = (V, E)$ to a low-dimensional embedding $z_v \in \mathbb{R}^d$ ($d \ll |V|$).

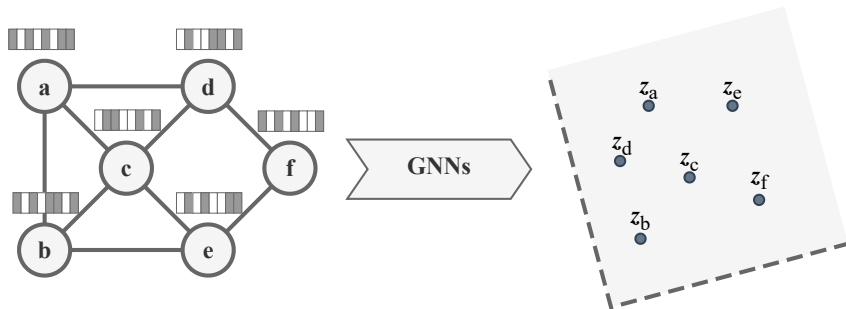


Figure 3: GNNs scheme.

Graph alignment problem

Let $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ be two graphs. The goal is to align their nodes through a mapping function $\pi : \mathbf{V}_1 \rightarrow \mathbf{V}_2$, where each node $v \in V_1$ is mapped to its corresponding node $\pi(v) \in V_2$.

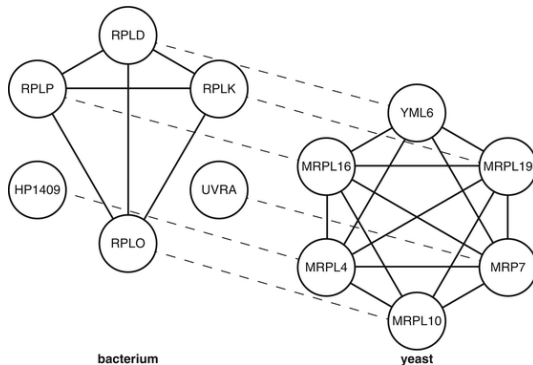


Figure 4: Node alignment.

Source: https://link.springer.com/rwe/10.1007/978-1-4419-9863-7_992

T-GAE: Transferable Graph Autoencoder for Network Alignment.
(He, J., Kanatsoulis, C., & Ribeiro, A., 2024)



Figure 5: T-GAE encoder.

Method

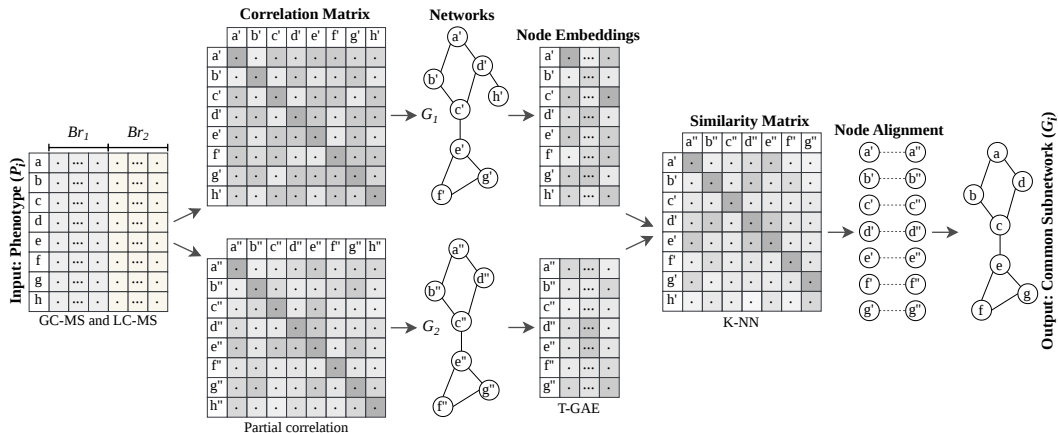


Figure 6: Our workflow.

- ① **Correlation estimation:** compute a partial correlation matrix.
- ② **Network construction:** build a metabolic network using a correlation threshold ($|\text{coef}| \geq 0.5$) and incorporate node attributes (R_t , M_z , and intensities).
- ③ **Graph representation learning:** learn node embeddings with the T-GAE architecture based on GIN and GINE layers.
- ④ **Network alignment:** identify metabolite correspondences using a KNN-based similarity matrix.

The experiments were conducted on a computer with 256 GB of RAM and 48 GB of VRAM. The tools used were Python and the PyTorch Geometric library.

Table 1: Datasets.

Dataset	Type network	#Nodes	#Edges	#Features	#Anchors/Pairs
Douban Online	Social	3 906	1 632	538	1 118
Douban Offline	Social	1 118	3 022	538	1 118
ACM	Citation	9 916	44 808	17	6 325
DBLP	Citation	9 872	39 561	17	6 325

Table 2: Dataset information.

Group	Subgroup	$ V $	$ E $	Density	Connected
<i>secoAmazonas</i>	1	2071	1 113 277	0.52	True
<i>secoAmazonas</i>	2	2071	1 085 089	0.51	True
<i>secoSanMartin</i>	1	2069	1 082 335	0.51	True
<i>secoSanMartin</i>	2	2071	1 103 292	0.51	True
<i>secoCusco</i>	1	2072	1 091 447	0.51	True
<i>secoCusco</i>	2	2072	1 086 182	0.51	True
<i>frescoCusco</i>	1	2071	1 114 859	0.52	True
<i>frescoCusco</i>	2	2072	1 089 225	0.51	True

Node embedding (T-GAE)

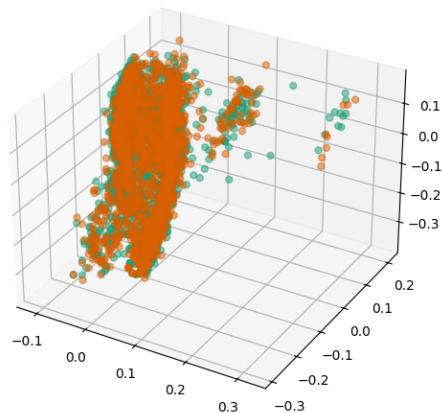


Figure 7: Node embedding (*secoSanMartin*).

Table 3: Node alignment (*secoSanMartin*).

ID 1	ID 2
13	13
19	252
27	378
31	1148
38	38
43	273
49	72
50	50
51	366
53	241
73	270
...	...
1825	1726

Table 4: Node filtered **133/2072** (*secoSanMartin*).

ID	Rt	Mz	Metabolite name
13	6.10	281.05	Unknown
38	6.16	182.10	Unknown
50	6.20	354.10	Unknown
128	6.47	119.20	Unknown
184	6.76	341.14	Unknown
261	7.23	371.06	Unknown
272	7.29	327.01	Unknown
314	7.61	57.02	DECANE; EI-B; MS
336	7.77	53.96	Unknown
339	7.81	328.01	Unknown
400	8.31	109.08	CITRONELLYL ACETATE...
402	8.31	109.13	BETA-HOMOCYCLOCITRAL...
407	8.37	401.11	Unknown

Thanks!
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<https://github.com/win7>

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